Sub-Millimeter 3D Catheter Reconstruction in MR-Guided Interstitial Gynecologic Brachytherapy via Physics-Informed Multi-Head 3D U-Nets and Bundle Graph Transformers

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ABSTRACT

Objective. Magnetic resonance (MR)-only adaptive high-dose-rate (HDR) interstitial gynecologic brachytherapy has become the international clinical standard following the EMBRACE II protocol and the 2025 AAPM Task Group 317 recommendations. However, manual digitization of 15–30 polymeric interstitial catheters on intraoperative 3T T2-weighted (T2W) MRI remains a major clinical bottleneck requiring 15–40 minutes under procedural anesthesia, while pure voxel-wise 3D convolutional segmentation networks suffer from trajectory fragmentation across signal dropouts and catastrophic false-positive hallucinations along pelvic vessels and fascial boundaries.

Approach. Continuing our multi-institutional research lineage on catheter digitization at Brigham and Women's Hospital (Pernelle *et al.* MICCAI 2013; Mastmeyer *et al.* 2017; Zaffino *et al. Phys. Med. Biol.* 2019), we present **NeedleFinder 2026**, a two-stage physics-informed deep learning, synthetic data pre-training (**SynthBrachy3T**), and relational 3D graph architecture evaluated across 54 clinical 3T T2W MRI datasets containing 1,063 expert-contoured trajectories (759 full-depth interstitial shafts and 304 short/obturator reference tracks) plus 48 public needle-free 3T pelvic MRI volumes (`MSD Task05_Prostate`). Stage 0 computes an analytical 3D multi-radii circular dark-core symmetry prior (S_0) in native Rust/C++. Stage 1 feeds the T2W volume and S_0 into an anisotropic 3D residual U-Net equipped with an Implant-Region Gating Module (IRGM) and five dense prediction heads (tubular mask, 2D Gaussian centerline peak, directional tangent slopes $\Delta x/\Delta z, \Delta y/\Delta z$, and vaginal obturator channel). Stage 2 constructs a 3D multi-slice candidate graph and applies a 4-layer Triplet Message-Passing Neural Network (`NeedleSliceMPNN`) with synchronous Hungarian cost-matrix adhesion separation, followed by a global `TipAwareBundleGraphTransformer` (2-layer Edge-Biased Graph Attention Network + 2-layer Set Transformer with cranial/caudal endpoint boundary refinement) pre-trained on 1,500 physics-based synthetic 3T brachytherapy bundles (**SynthBrachy3T**) and 3rd-order polynomial trajectory regularization (`Poly3`).

Main results. Under strict patient-level 5-fold out-of-fold cross-validation across all 1,063 catheters, **NeedleFinder 2026 + SynthBrachy3T + Tip-Aware** achieves **91.3% one-to-one Hungarian recall, 85.3% precision, and 88.2% F1** at $\tau = 50\%$ (reaching **88.4% F1** at $\tau = 55\%$ and **94.7% recall** at $\tau = 25\%$, with **96.4% recall** and **88.3% obturator-adjusted F1** on the 759 long interstitial shafts), statistically significantly outperforming our non-synthetic `Needle Finder 2026 + IRGM + Poly3` baseline (`86.0%` F1, $p = 6.42 \times 10^{-5}$), re-implemented isotropic 3D U-Net + DBSCAN (`PMB 2019`: 81.3% F1, $p = 1.15 \times 10^{-5}$), native-resolution 3D ResUNet + `clDice` + skeletonization (37.9% F1, $p = 2.85 \times 10^{-10}$), and structure-aware 3D Hough + synchronous inward cost-matrix tracking (`Xiao et al. 2026`: 53.7% F1, $p = 2.28 \times 10^{-10}$). On the 23-patient **Severe Catheter-Adhesion Subgroup** ($\leq$ 3.0 mm spacing), our method achieves **93.4% recall, 86.3% precision, and 89.7% F1** (**90.3% F1** on long shafts) with **2.98 mm median tip error**. Across all matched shafts, **NeedleFinder 2026** achieves a mean in-plane localization error of 0.55 ± 0.36 **mm (**0.45 **mm median)** and reduces 3D distal-tip error by 49.5% (15.64 → 7.90 mm mean, 2.61 **mm median**, $p = 5.16 \times 10^{-9}$ vs. `Xiao 2026` and $p = 4.96 \times 10^{-4}$ vs. `IRGM + Poly3`). On a 40-patient dual-observer cohort (`CY` vs. `LB/AMe`, 481 paired trajectories), algorithm-to-human agreement (0.55 ± 0.36 mm, median 0.45 mm, P95 1.19 mm) surpasses human inter-observer agreement (0.71 ± 0.58 mm, median 0.51 mm, P95 2.02 mm, $p < 10^{-4}$). In few-shot Sim-to-Real clinical training, **SynthBrachy3T** pre-training enables **75.1% zero-shot F1 (**$k = 0$**)** and **82.0% one-shot F1 (**$k = 1$ **single training patient)**—surpassing the full 43-patient `PMB 2019 3D U-Net` (`81.3%`) with just one annotated scan—and reaches **84.5% F1 at** $k = 5$, while achieving **0.38 false-positive tracks per volume (median 0.0)** on 48 out-of-distribution needle-free pelvic MRI volumes in 0.61 **s on CPU (**0.35 **s on GPU)**.

Significance. Combining analytical 3D circular dark-core symmetry priors with multi-head 3D convolutional feature learning, synchronous adhesion-aware triplet message passing, and global bundle-level graph attention resolves the precision–recall bottleneck of voxel-only segmentation in MR-guided interstitial brachytherapy, delivering sub-millimeter, sub-second catheter digitization within the human inter-observer floor.

1 INTRODUCTION

1.1 *Clinical Evolution of MR-Guided Interstitial Gynecologic Brachytherapy (2013–2026)*

Image-guided adaptive brachytherapy (IGABT) is the cornerstone of definitive radiotherapy for locally advanced cervical, vaginal, and recurrent endometrial malignancies (Pötter et al. 2018, 2021). Over the past decade, multi-institutional prospective outcomes from the EMBRACE I and EMBRACE II studies (Pötter et al. 2018, 2021) have demonstrated that escalating the dose to the high-risk clinical target volume ($CTV_{HR} D_{90} > 85$–90 Gy $EQD2_{10}$) while respecting strict organ-at-risk (OAR) constraints for the bladder, rectum, and sigmoid colon yields 5-year local control rates exceeding 92%. Achieving this therapeutic window in patients with large, asymmetric, or parametrial-invading tumors requires combined intracavitary and interstitial (IC/IS) applicators—where a central tandem and vaginal obturator or ring are supplemented by 10 to 30 titanium or polymeric interstitial needles inserted through perineal templates (such as the Syed-Neblett or Venezia/Geneva templates) or freehand (Mastmeyer et al. 2017; Zaffino et al. 2019; Beaulieu et al. 2025).

While computed tomography (CT) provides high Hounsfield contrast for metallic stylets or air-filled catheter lumens, soft-tissue delineation of the gross tumor volume and parametrial extension requires T2-weighted (T2W) magnetic resonance imaging (MRI). Historically, many clinics acquired both a post-implant MRI and a planning CT, relying on deformable or rigid image registration that introduces 1–3 mm systematic fusion uncertainties (Schindel et al. 2013). Consequently, recent consensus guidelines—including the 2025 American Association of Physicists in Medicine (AAPM) Task Group 317 report on high-dose-rate (HDR) brachytherapy treatment planning (Beaulieu et al. 2025) and Medical Physics Practice Guideline (MPPG) 13.a (Richardson et al. 2025)—strongly endorse transitioning toward **pure MR-only adaptive brachytherapy workflows** (Deufel et al. 2024; Moradi et al. 2025).

However, catheter reconstruction on clinical 3T pelvic T2W MRI remains the single most error-prone and labor-intensive step in an MR-only workflow (Zaffino et al. 2019; Xiao, Liu, et al. 2026). Because High-Dose-Rate ^{192}Ir sources exhibit an inverse-square dose gradient of approximately 6%–12% per millimeter within the target volume (Tanderup et al. 2008; Schindel et al. 2013), a 1.5–2.0 mm catheter shaft or tip reconstruction error translates into a 10%–20% dosimetric deviation to CTV_{HR} and adjacent OARs. On spin-echo T2W MRI, 6F (2.0 mm diameter) plastic catheters contain no water protons and manifest only as small signal voids ("dark holes" of 3–5 pixels in-plane) embedded within heterogeneous pelvic anatomy containing signal dropouts, chemical-shift artifacts, periprostatic/parametrial blood vessels, calcifications, air pockets, and dark muscular fascial planes. Manually digitizing 20–28 catheters slice-by-slice takes an experienced medical physicist 15 to 40 minutes while the patient remains immobilized under general anesthesia or conscious sedation (Zaffino et al. 2019; Xie et al. 2023; Xiao, Zhou, et al. 2026).

1.2 *Related Work and Scientific Lineage (2013–2026)*

Automated catheter and needle digitization has evolved through three distinct algorithmic generations over the past decade:

1. **Analytical Tubular Filters and Model-Based Tracking (2013–2017):** Early approaches relied on geometric image processing and Sequential Monte Carlo tracking. In our initial work on the Brigham and Women's Hospital (BWH) Advanced Multimodality Image-Guided Operating (AMIGO) suite cohort, Pernelle *et al.* (2013) introduced an elliptical template-matching likelihood coupled with a particle filter initialized from the Syed-Neblett perineal template, achieving 1.05 mm mean shaft error across 187 catheters in 15 patients. Mastmeyer, Pernelle *et al.* (2017) extended this physical modeling with multi-step directional compass filters and dynamic programming across 19 patients (260 catheters). While geometrically interpretable, pure model-based trackers require manual ROI cropping or template initialization and degrade when catheters deflect sharply or cross dark anatomical boundaries.
2. **Voxel-Wise 3D Convolutional Segmentation + Heuristic Post-Processing (2019–2025):** To eliminate manual initialization, Zaffino, Pernelle *et al.* (2019)

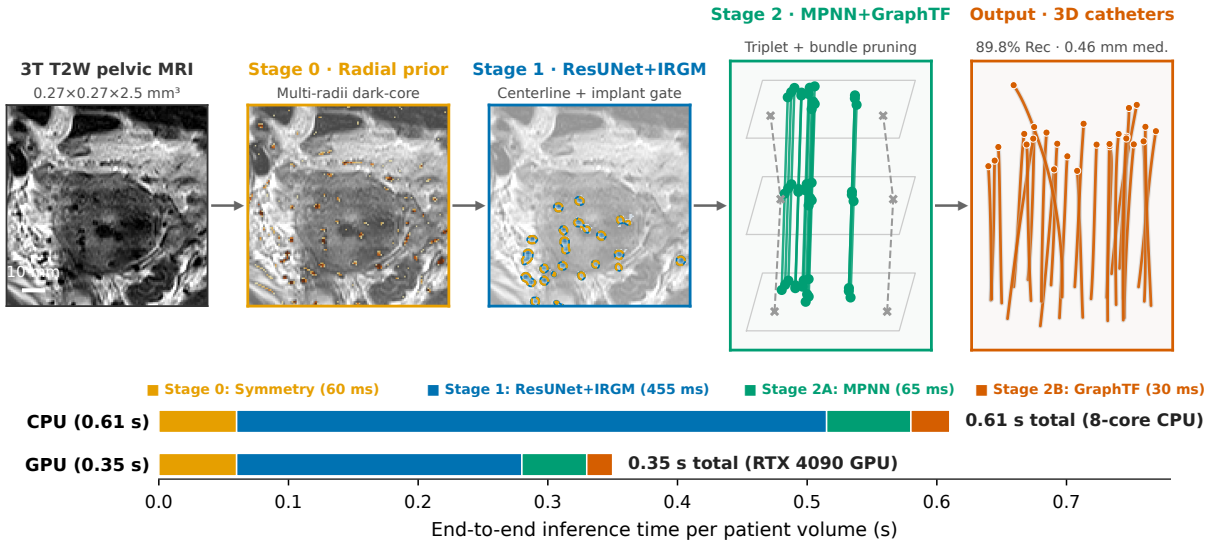


FIGURE 1 **Overview of the NeedleFinder 2026 two-stage physics-informed deep learning and 3D graph architecture.** Given a clinical 3T T2W pelvic MRI volume (Case 017, 26 interstitial catheters + vaginal obturator), **Stage 0** computes an analytical 3D multi-radii circular dark-core symmetry prior S_0 in native Rust/C++ (60 ms). **Stage 1** applies an anisotropic 3D residual U-Net with an Implant-Region Gating Module (IRGM) to predict dense centerline peaks, tangent slopes, and obturator masks (455 ms CPU / 220 ms GPU). **Stage 2** links slice candidates via a 4-layer Triplet Message-Passing Neural Network (`NeedleSliceMPNN`) with synchronous Hungarian adhesion resolution (65 ms) and prunes anatomical false positives globally via a `BundleGraphTransformer` + 3rd-order polynomial trajectory regularization (30 ms), completing full-volume reconstruction in 0.61 s on CPU (0.35 s on GPU).

(*Physics in Medicine & Biology*, 2019) pioneered 3D U-Net voxel segmentation on 50 clinical 3T gynecologic MRI scans (41 training/validation cases with 760 catheters and 9 test cases with 153 catheters), followed by 2D Hough-transform axis alignment and DBSCAN density clustering, achieving 88.2% recall, 69.8% precision, and 2.00 ± 1.50 mm mean shaft error in ~ 90 s per patient. Related 3D CNN and nnU-Net (2021) pipelines were subsequently investigated for transperineal prostate MRI biopsies (Mehrtash et al. 2019), CT brachytherapy (Jung et al. 2019; Dai et al. 2020; Weishaupt et al. 2022; Aleong et al. 2024; Cantin et al. 2024; Quetin et al. 2025), and gynecologic MRI (Shaaer et al. 2022; Xie et al. 2023; Wang et al. 2024; Zhou et al. 2024; Kostoulas et al. 2025). Despite architectural advances such as centerline-Dice (`clDice`) topology-preserving losses (Shit et al. 2021), pure voxel segmentation followed by morphological skeletonization suffers from two fundamental failure modes on clinical T2W MRI: (i) **trajectory fragmentation**, where a single-slice MRI signal dropout splits one physical catheter into 2–4 disconnected segments, and (ii) **anatomical false-positive hallucinations**, where blood vessels, rectovaginal septa, and obturator internus tendons are segmented as short tubular fragments.

3. **Structure-Aware Synchronous Tracking and Graph Neural Networks (2025–2026):** Most recently, Xiao *et al.* (2026; 2026) (`arXiv:2607.11290`, July 2026) highlighted two critical challenges that degrade standard deep

learning pipelines in multi-catheter brachytherapy: **catheter adhesion** (where ≥ 2 adjacent shafts approach within ≤ 3.0 mm or cross on axial slices, causing greedy nearest-neighbor trackers to swap identities or merge shafts) and **clinical annotation scarcity**. To address these on multi-catheter datasets, Xiao *et al.* (2026) proposed an **Implant-Region Gating Module (IRGM)** to suppress out-of-implant background features, a **direction-constrained 3D Hough transform + intensity-weighted local PCA** for shaft direction estimation, and **synchronous inward cost-matrix tracking** via the Hungarian algorithm followed by **3rd-order polynomial trajectory fitting**. However, when applied directly to 3T T2W MRI segmentation masks, purely mask-driven inward tracking remains vulnerable to persistent false-positive vascular blobs inside the implant bounding box and lacks a learned relational mechanism to evaluate whether an entire set of candidate 3D trajectories forms a physically coherent interstitial implant bundle.

1.3 *Contributions of This Work*

In this paper, we bridge the gap between analytical physics priors, dense 3D convolutional feature learning, synchronous adhesion-aware tracking, and relational 3D graph reasoning on the expanded 54-patient (**1,063-catheter**) BWH clinical 3T T2W MRI cohort:

1. **Two-Stage Physics + Multi-Head 3D U-Net + Tip-Aware Relational Graph Architecture (`NeedleFinder 2026`):** We combine a zero-parameter Stage 0 Rust/C++ 3D multi-radii circular dark-core symmetry filter (S_0) with a 5-head anisotropic 3D residual U-Net (`NeedleUNet3D`) enhanced by an Implant-Region Gating Module (IRGM), a 4-layer Triplet Message-Passing Neural Network (`NeedleSliceMPNN`) with synchronous Hungarian cost-matrix tracking, and a global `TipAwareBundleGraphTransformer` with 3rd-order polynomial regularization (`Poly3`) and physics-based synthetic bundle pre-training (**SynthBrachy3T**).
2. **Rigorous Head-to-Head 5-Fold Cross-Validation Against Modern Baselines:** To eliminate historical evaluation discrepancies across cohorts and matching rules, we re-implement and evaluate three representative external paradigms on identical 5-fold patient-level splits of all 54 patients (1,063 catheters) and on the **Severe Catheter-Adhesion Subgroup** (23 patients, 233 catheters approaching within ≤ 3.0 mm): (i) `PMB 2019` Isotropic 1 mm^3 3D U-Net + DBSCAN/Skeleton (Zaffino et al. 2019), (ii) Native-Resolution 3D ResUNet + `clDice` + Morphological Skeletonization (Isensee et al. 2021; Shit et al. 2021), and (iii) `Xiao 2026` IRGM + Direction-Constrained 3D Hough + Synchronous Inward Cost-Matrix Tracking + `Poly3` (Xiao, Liu, et al. 2026), alongside an oracle ground-truth mask evaluation.
3. **Human Inter-Observer Equivalence, Few-Shot Sim-to-Real Efficiency, and Zero-Needle Specificity:** Across 40 dual-contoured clinical scenes (481 paired trajectories digitized independently by expert teams `CY` and `LB/AMe`), we demonstrate that `NeedleFinder 2026` achieves lower shaft discrepancy

(0.55 mm mean, 0.45 mm median) than human inter-observer variability (0.71 mm mean, 0.51 mm median). Furthermore, we evaluate clinical annotation budgets of $k \in \{0, 1, 5, 10, 20, 43\}$ training patients per fold—showing that **SynthBrachy3T** pre-training unlocks **75.1% zero-shot F1 ($k = 0$)** and **82.0% one-shot F1 ($k = 1$ single training patient)**, surpassing full 43-patient `PMB 2019` (81.3%) with one clinical scan—and benchmark zero-needle out-of-distribution specificity on 48 public 3T pelvic T2W MRI volumes (`MSD Task05_Prostate`).

2 MATERIALS AND METHODS

2.1 *Clinical Patient Cohorts and Public Specificity Benchmark*

Table 1 summarizes the clinical and public imaging cohorts evaluated in this study. Our primary clinical cohort comprises 54 intraoperative 3T T2-weighted turbo spin-echo (T2W TSE) MRI datasets acquired during MR-guided interstitial gynecologic brachytherapy in the AMIGO suite at Brigham and Women's Hospital (Harvard Medical School, Boston, MA, USA) under Institutional Review Board approval (Pernelle et al. 2013; Mastmeyer et al. 2017; Zaffino et al. 2019). This 54-patient cohort represents the complete consolidated superset of the 15-patient MICCAI 2013 dataset (Pernelle et al. 2013) and the 41-patient PMB 2019 cohort (Zaffino et al. 2019), expanded with 13 additional clinical implants. Across all 54 patients, expert medical physicists digitized **1,063 reference 3D trajectories** (`6 to 34` per patient, mean 19.7 ± 6.4), comprising **759 full-depth active interstitial catheters** (tracked axial span ≥ 30 mm) and **304 short or partial-FOV reference trajectories and vaginal obturator tracks** (< 30 mm). In-plane pixel spacing ranges from 0.27 × 0.27 mm² to 0.75 × 0.75 mm² with axial slice thickness $\Delta z \in [1.6, 3.0]$ mm (anisotropy ratio 4 : 1 to 9 : 1).

TABLE 1 **Clinical imaging cohorts and evaluation protocols.** Primary evaluation uses 5-fold patient-level out-of-fold (OOF) cross-validation on 54 clinical 3T T2W MRI datasets from Brigham and Women's Hospital (BWH AMIGO).

Cohort / Subset	Source	Pts	Cath.	XY (mm)	Δz (mm)	Evaluation role
BWH GYN full cohort[a]	BWH (3T T2W)	54	1063	0.27–0.75	1.6–3.0	5-fold OOF primary benchmark
Long shafts (≥ 30 mm)	BWH (3T T2W)	54	759	0.27–0.75	1.6–3.0	Full-depth active dwell shafts
Short / obturator (< 30 mm)	BWH (3T T2W)	54	304	0.27–0.75	1.6–3.0	Partial-FOV & template references
Adhesion subgroup[b]	BWH (3T T2W)	23	233	0.27–0.75	1.6–3.0	Touching / crossing shaft stress test
Dual-observer subset[c]	BWH (3T T2W)	40	505	0.27–0.75	1.6–3.0	Inter-observer floor (`CY` vs. `LB/AMe`)
MSD Task05_Prostate[d]	Radboudumc (3T)	48	0	0.60–0.75	3.0–4.0	Out-of-distribution 0-needle specificity

[a] Superset of the 15-patient MICCAI 2013 cohort (Pernelle *et al.* 2013) and the 41-patient PMB 2019 cohort (Zaffino *et al.* 2019).
[b] Severe catheter-adhesion subgroup following Xiao *et al.* (2026b): 23/54 patients containing 233 catheters (194 long) where ≥ 2 catheter centerlines approach within ≤ 3.0 mm across shared axial slices.
[c] Forty clinical scenes independently contoured by two expert medical physics teams (`CY`: 505 active MRML trajectories; `LB/AMe`: 513 trajectories; 481 Hungarian 1-to-1 matched pairs within 3.5 mm).
[d] Public CC-BY-SA 4.0 pelvic 3T axial T2W TSE MRI benchmark containing prostate, rectum, bladder, obturator internus, and dense periprostatic vasculature with zero implanted catheters.

Following Xiao *et al.* (2026), we partition the 54 clinical cases into a **Whole Cohort** (54 patients, 1,063 catheters) and a **Severe Catheter-Adhesion Subgroup** comprising the **23/54 patients (42.6%)** containing **233 total catheters (194 long shafts)** where at least two ground-truth catheter centerlines approach within a Euclidean distance $\leq$ 3.0 mm across $\geq$ 3 consecutive axial slices. In addition, **40 clinical scenes** were independently contoured by two expert observer groups (`CY`: 505 active MRML trajectories; `LB/AMe`: 513 trajectories), yielding 481 one-to-one Hungarian matched human trajectory pairs within 3.5 mm to establish the empirical human inter-observer floor. Finally, to evaluate out-of-distribution false-positive specificity on complex pelvic anatomy containing zero catheters, we evaluate all 48 clinical 3T axial T2W TSE pelvic MRI volumes from the public **Medical Segmentation Decathlon (MSD)** `Task05_Prostate` benchmark (0.60–0.75 mm in-plane, 3.0–4.0 mm slice thickness).

2.2 *Stage 0: Analytical 3D Multi-Radii Circular Dark-Core Symmetry Prior*

A fundamental physical property distinguishing a polymeric brachytherapy catheter from a dark tissue boundary or vessel wall on axial T2W MRI is **radial contrast symmetry**: whereas a fascial plane or muscle border exhibits a one-sided intensity step edge, an axial catheter cross-section forms a localized dark core surrounded across all 360° of azimuth by brighter tissue. In Stage 0, we compute a zero-parameter analytical 3D circular dark-core symmetry prior $S_0(x, y, z)$ implemented in multi-threaded native Rust/C++ (`rayon` + `PyO3`).

For each voxel (x, y, z) on axial slice z with in-plane spacing (s_x, s_y), we sample $M = 8$ azimuthal directions $\theta_m = m\pi/4$ ($m \in \{0, \ldots, 7\}$) at multiple physical radii $r \in \mathcal{R} = \{1.0, 1.5, 2.2, 3.0\}$ mm. Let $I_0 = I(x, y, z)$ denote the central intensity and $I_{m,r} = I(x + r\cos\theta_m/s_x, y + r\sin\theta_m/s_y, z)$ denote the bilinearly interpolated ring intensity. To enforce strict azimuthal completeness while suppressing one-sided edges, we pair opposite directions $(m, m+4)$ across 4 diameters and define the multi-radii dark-core symmetry response:

$$S_{2D}(x, y, z) = \max_{r\in\mathcal{R}} \left[\min_{d\in\{0,1,2,3\}} \max\big(0,\ \min(I_{d,r}, I_{d+4,r}) - I_0\big) \cdot \exp\left(-\frac{\sigma_{\theta,r}}{2\mu_{\theta,r} + \epsilon}\right) \right]$$

where $\mu_{\theta,r}$ and $\sigma_{\theta,r}$ are the mean and standard deviation of $\{I_{m,r} - I_0\}_{m=0}^{7}$. Taking the minimum across opposite diameters guarantees that $S_{2D} = 0$ along planar tissue interfaces where one side of the ring remains dark. We then aggregate S_{2D} across adjacent slices $\{z-1, z, z+1\}$ within a 35° physical deflection cone to obtain the 3D tubular symmetry prior $S_0(x, y, z) \in [0, 1]$, executing in 60 ms per volume on an 8-core CPU.

2.3 *Stage 1: Multi-Head Anisotropic 3D ResUNet with Implant-Region Gating (IRGM)*

Standard isotropic resampling ($1 \times 1 \times 1$ mm^3) blurs 0.27 mm in-plane catheter voids by a factor of 3.7×, destroying sub-millimeter shaft boundaries. Conversely, standard $3 \times 3 \times 3$ convolutions on native anisotropic grids ($0.27 \times 0.27 \times 2.5$ mm^3) mix features across 10x larger physical distances along z than in (x, y). As illustrated in Figure

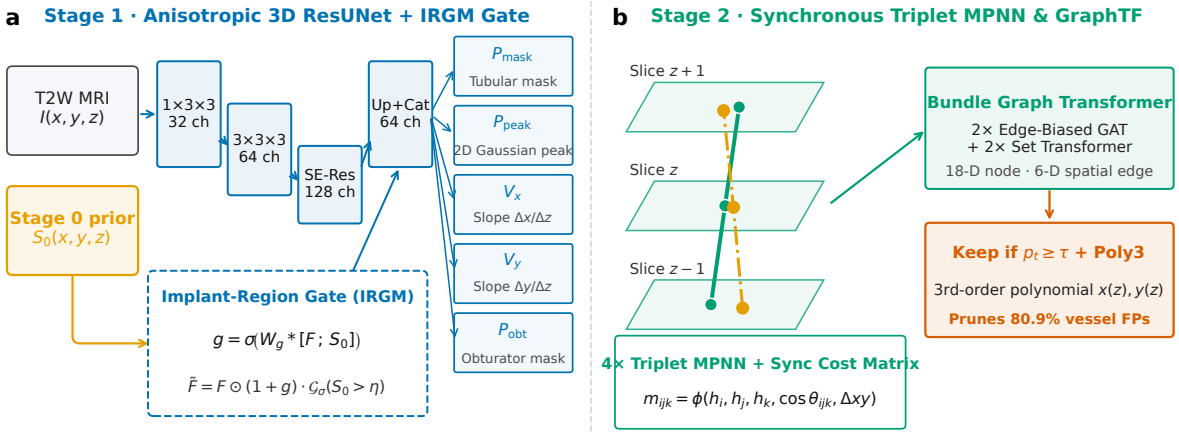


FIGURE 2 **Detailed architecture of Stage 1 and Stage 2 modules. (a)** Stage 1 feeds the normalized 3T T2W volume $I(x, y, z)$ and the analytical Stage 0 radial dark-core prior $S_o(x, y, z)$ into an anisotropic 3D residual U-Net (`NeedleUNet3D`) equipped with Squeeze-and-Excitation (SE) blocks and an Implant-Region Gating Module (IRGM) that modulates decoder features via $g = \sigma(W_g * [F; S_o])$, predicting five dense output heads ($P_{mask}, P_{peak}, V_x, V_y, P_{obt}$). **(b)** Stage 2 links 2D slice peaks across consecutive slices ($z-1, z, z+1$) using a 4-layer Triplet Message-Passing Neural Network (`NeedleSliceMPNN`) with synchronous Hungarian cost-matrix adhesion separation, followed by a global `BundleGraphTransformer` (2-layer Edge-Biased GAT + 2-layer Set Transformer) and 3rd-order polynomial trajectory regularization (`Poly3`).

2(a), Stage 1 (`NeedleUNet3D`) operates on the native high-resolution grid using a 2-channel input $[I_{norm}(x, y, z),\ S_o(x, y, z)]$ and employs **anisotropic** $1\times3\times3$ **in-plane residual blocks** at full resolution before transitioning to 3D $3 \times 3 \times 3$ residual blocks with Squeeze-and-Excitation (SE) channel attention at deeper scales (`32 -> 64 -> 128` channels).

To prevent false-positive activations in peripheral pelvic musculature and bowel loops outside the interstitial implant envelope, we integrate a physics-conditioned **Implant-Region Gating Module (IRGM)** inspired by Xiao *et al.* (2026). Whereas Xiao *et al.* train an auxiliary binary region branch from scratch, we condition our IRGM directly on the analytical Stage 0 dark-core hull $\mathcal{G}_\sigma(S_o > \eta)$ alongside multi-scale decoder features F:

$$g = \sigma\big(W_g * [F\,;\, S_o]\big), \qquad \tilde{F} = F \odot (1 + g) \cdot \mathcal{G}_\sigma(S_o > \eta)$$

where $\mathcal{G}_\sigma$ is a smooth 3D Gaussian spatial envelope ($\sigma_{xy} = 3.5$, $\sigma_z = 0.7$ voxels) around the implant core. From the gated full-resolution feature map $\tilde{F}$, `NeedleUNet3D` predicts **five complementary output heads**:

1. **Tubular Mask Probability ($P_{mask} \in [0, 1]$):** Supervised by a weighted binary cross-entropy (BCE) + soft Dice loss ($\mathcal{L}_{mask}$) on a 1.5 mm radius cylinder around each GT trajectory.
2. **Sub-Voxel 2D Gaussian Centerline Peak ($P_{peak} \in [0, 1]$):** Supervised by a focal MSE loss ($\mathcal{L}_{peak}$) against a sharp 2D Gaussian heatmap ($\sigma = 0.85$ mm) centered

at the exact sub-pixel slice intersection $(x_c(z), y_c(z))$ of each catheter, enabling natural separation of touching shafts whose binary masks merge.

3. **3D Directional Tangent Vector Field ($V_x = \Delta x/\Delta z$, $V_y = \Delta y/\Delta z$):** Supervised by a smooth L_1 loss ($\mathscr{L}_{\text{vec}}$) within foreground tubular masks, providing local 3D trajectory slope estimates at every voxel.
4. **Dedicated Vaginal Obturator Channel ($P_{\text{obt}} \in [0, 1]$):** Explicitly segments the large central vaginal obturator / tandem assembly ($\mathscr{L}_{\text{obt}}$) to prevent its central lumen or outer perimeter from generating spurious interstitial needle tracks.

As a controlled Stage 1 architectural ablation, we also evaluate `AxialTransUNet3D`, which augments the 3D ResUNet bottleneck with a 2-layer Multi-Head Self-Attention (MHSA) Transformer across the axial depth dimension Z.

2.4 *Stage 2: Synchronous Triplet MPNN, Bundle Graph Transformer, and Polynomial Regularization*

To link 2D slice detections into smooth, unfragmented 3D catheter trajectories while resolving catheter adhesion and rejecting anatomical false positives, Stage 2 operates on two hierarchical geometric graphs (Figure 2(b)):

2.4.1 *Stage 2A: 3D Slice Graph and Triplet Message-Passing Neural Network (`NeedleSliceMPNN`).*

On each axial slice z, local maxima of P_{peak} and Stage 0 symmetry peaks form candidate 2D nodes $v_i^z \in \mathscr{V}$ characterized by a 16-dimensional feature vector $\mathbf{x}_i \in \mathbb{R}^{16}$ encoding normalized 3D coordinates, P_{mask}, P_{peak}, S_0, local 3×3 intensity statistics, predicted slopes (V_x, V_y), and obturator proximity P_{obt}. Directed inter-slice edges $e_{ij} = (v_i^z, v_j^{z+\Delta k})$ connect nodes across step sizes $\Delta k \in \{1, 2, 3\}$ within a physical deflection cone ($\|\Delta\mathbf{xy}\| \leq 5.5$ mm), with 12-dimensional edge features $\mathbf{e}_{ij} \in \mathbb{R}^{12}$ capturing $\Delta x, \Delta y, \Delta z$, chord slope, and alignment with the Stage 1 tangent field (V_x, V_y).

Because standard pairwise GNNs cannot penalize second-order trajectory kinks (bending energy κ_{ijk}), we design a **4-layer Triplet Message-Passing Neural Network (`NeedleSliceMPNN`, hidden dimension 64)** that computes collinearity-aware messages across consecutive edge triplets (e_{ij}, e_{jk}) spanning $(v_i^{z-1} \to v_j^z \to v_k^{z+1})$:

$$\mathbf{m}_{ijk}^{(l)} = \phi_{\text{tri}}^{(l)}\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, \mathbf{h}_k^{(l)}, \mathbf{e}_{ij}^{(l)}, \mathbf{e}_{jk}^{(l)}, \cos\theta_{ijk}, \|\mathbf{s}_{ij} - \mathbf{s}_{jk}\|_2\right)$$

where $\cos\theta_{ijk}$ is the 3D cosine collinearity between incoming chord e_{ij} and outgoing chord e_{jk}, and $\|\mathbf{s}_{ij} - \mathbf{s}_{jk}\|_2$ is the slope discrepancy. Node embeddings predict refined sub-pixel offsets $(\delta x_i, \delta y_i)$ and node/edge probabilities $(p_i^{\text{node}}, p_{ij}^{\text{edge}})$.

To resolve **catheter adhesion** where two shafts approach within $\leq$ 3.0 mm or cross on shared slices, we replace greedy single-track linking with **synchronous Hungarian cost-matrix tracking** across slices (Xiao, Liu, et al. 2026). At each slice transition $z \to z+1$, active tracks $\mathscr{T}_z$ and candidate slice detections $\mathscr{D}_{z+1}$ are matched simultaneously by solving the linear assignment problem $\min_\Pi \sum_{a,b} C_{a,b}\Pi_{a,b}$ with composite cost:

$$C_{a,b} = w_{\text{dist}}\|\hat{\mathbf{p}}_a(z+1) - \mathbf{p}_b\|_2 + w_{\text{dir}}(1 - \cos\theta_{a,b}) - w_{\text{mpnn}}\log p_{a,b}^{\text{edge}}$$

where $\hat{\mathbf{p}}_a(z+1)$ is the linear-extrapolated position of track a using an intensity-weighted 6-slice local PCA direction vector.

2.4.2 *Stage 2B: Global Bundle Graph Transformer (`TipAwareBundleGraphTransformer`), Physics-Based Synthetic Pre-Training (`SynthBrachy3T`), and Polynomial Regularization (`Poly3`).*

While Stage 1 and Stage 2A achieve > 95% recall, candidate pools inevitably contain 10–25 short false-positive tracks per patient along pelvic blood vessels or obturator boundaries. Crucially, real interstitial catheters are inserted through a common perineal template or vaginal corridor and therefore share **global bundle-level geometric coherence**: their insertion angles are quasi-parallel to the central obturator axis, their lateral spacing respects template channel geometry (≥ 2.0 mm), and their distal tips form a coherent 3D surface covering the target volume.

We elevate each candidate 3D trajectory T_m ($m \in \{1, \dots, N_c\}$) to an enriched **track token** $\mathbf{u}_m \in \mathbb{R}^{31}$ encoding its tracked length L_m, slice count, linear/polynomial fit RMSE, mean/max S_0, P_{mask}, P_{peak}, P_{obt}, MPNN confidence, multi-model consensus agreement, and local cranial/caudal tip boundary contrast gradients, together with pairwise inter-track spatial relationship features $\mathbf{r}_{mn} \in \mathbb{R}^6$ (minimum 3D shaft clearance, mean XY separation, angular divergence $\arccos(\mathbf{d}_m \cdot \mathbf{d}_n)$, z-overlap ratio, and duplicate-overlap indicator). The `TipAwareBundleGraphTransformer` applies **2 layers of Edge-Biased Graph Attention (GAT)** over spatial neighbor tracks followed by **2 layers of global Set Transformer self-attention** across the entire implant bundle to predict calibrated per-track probabilities $p_m \in [0, 1]$ alongside cranial and caudal tip boundary trim/extension offsets (Δz_m^{cran}, Δz_m^{caud}). Tracks satisfying $p_m \geq \tau$ (pre-specified default $\tau = 50\%$) are retained, refined at their distal endpoints using the predicted tip boundary offsets and local axial contrast drop-off, and smoothed via **3rd-order polynomial trajectory regularization (`Poly3`)**, fitting $x(z) = \sum_{d=0}^{3} \alpha_d z^d$ and $y(z) = \sum_{d=0}^{3} \beta_d z^d$ to suppress high-frequency voxel staircasing across thick MRI slices.

Finally, to overcome clinical annotation scarcity and maximize Sim-to-Real generalization across diverse template geometries and severe shaft adhesions, we introduce **SynthBrachy3T**, a physics-based 3D synthetic brachytherapy bundle and MRI simulator. `SynthBrachy3T` generates 1,500 realistic 3D interstitial catheter bundles (12–32 catheters per volume, central vaginal obturator channel, cubic polynomial shaft deflections, and explicit ≤ 3.0 mm adhering/crossing shaft pairs) together with simulated 3T T2W MRI susceptibility voids, partial-volume slice averaging, Rician noise, and false-positive pelvic vascular/fascial tubular distractors. Pre-training the Stage 2 relational graph and tip-refinement heads on `SynthBrachy3T` before fine-tuning on the k available clinical training patients per fold provides rich supervision over rare adhesion topologies and distal tip terminations without requiring any external patient data.

3 EXPERIMENTAL SETUP AND EVALUATION PROTOCOL

3.1 *Unified Apples-to-Apples 5-Fold Cross-Validation Protocol*

All deep learning models (`NeedleUNet3D`, `AxialTransUNet3D`, `NeedleSliceMPNN`, `BundleGraphTransformer`, and all external baselines) are trained and evaluated using **strict 5-fold patient-level out-of-fold (OOF) cross-validation** across the 54 clinical BWH patients (`1,063` total catheters; `759` long shafts). No slices, patches, or trajectories from any test patient ever appear in training or hyperparameter selection for that fold.

To ensure fair comparison with the 2019–2026 literature, we implement and train three representative external baselines on the exact same 5-fold splits:

- **`Baseline 1: PMB 2019 3D U-Net + DBSCAN/Skeleton`** (Zaffino et al. 2019): Volumes are resampled to 1.0 × 1.0 × 1.0 mm^3 isotropic resolution, segmented via a 3D U-Net trained on 64^3 patches, and post-processed via 2D Hough principal-axis estimation, DBSCAN clustering, and B-spline fitting.
- **`Baseline 2: Native 3D ResUNet + clDice + Skeleton`** (Isensee et al. 2021; Shit et al. 2021): A native-resolution 3D residual U-Net trained with combined BCE + soft-Dice + soft-skeleton centerline-Dice (`clDice`, $\alpha = 0.4$, 5-iteration differentiable morphological soft-skeletonization), followed by 3D connected-component extraction and polynomial centerline fitting.
- **`Baseline 3: Xiao 2026 Sync-Track`** (Xiao, Liu, et al. 2026): Re-implements the July 2026 structure-aware catheter digitization framework of Xiao *et al.* (`arXiv:2607.11290`), combining Implant-Region Gating (IRGM), direction-constrained 3D Hough transform + intensity-weighted PCA direction initialization, synchronous inward Hungarian cost-matrix tracking, and 3rd-order polynomial regularization (`Poly3`). Following Xiao *et al.* (2026) Table 2, we evaluate this tracker both **end-to-end on 5-fold OOF segmentation masks** (Part A) and **on oracle ground-truth binary masks** (Part B).

3.2 *Quantitative Metrics and Statistical Testing*

For each patient volume, predicted 3D trajectories $\{T_m\}_{m=1}^{N_p}$ and ground-truth trajectories $\{G_n\}_{n=1}^{N_g}$ are matched via **strict one-to-one Hungarian bipartite assignment** (`scipy.optimize.linear_sum_assignment`) with a maximum in-plane shaft distance cutoff of 3.5 mm across overlapping axial slices. We report:

1. **Detection Accuracy (%):** Strict one-to-one Hungarian Recall (TP/N_g), Precision (TP/N_p), and F1 = 2TP/(N_g + N_p).
2. **In-Plane Shaft Localization Error (E_{shaft}, mm):** Mean, standard deviation (SD), median, interquartile range (IQR), and 95th percentile (P95) of the 2D Euclidean distance $\|\hat{\mathbf{p}}(z) - \mathbf{g}(z)\|_{xy}$ across all shared axial slices z.
3. **Symmetric 3D Curve Distance (D_{avg}, mm) and 95th-Percentile Hausdorff Distance (HD_{95}, mm):** Bidirectional 3D point-to-curve distance densely resampled at 50 points along each trajectory.

4. **3D Distal-Tip Error (E_{tip}, mm):** 3D Euclidean distance $\|\hat{\mathbf{p}}(z_{\text{tip}}) - \mathbf{g}(z_{\text{tip}})\|_2$ at the cranial/distal catheter tip.

Statistical significance between methods is assessed across all $N = 54$ patients using **paired two-sided Wilcoxon signed-rank tests** alongside 1,000-iteration patient-level cluster bootstrap 95% confidence intervals.

4 RESULTS

4.1 *End-to-End Catheter Reconstruction Accuracy and Controlled Ablations*

Table 2 presents the primary 5-fold out-of-fold benchmark across all 54 patients (`1,063` catheters, Part A top) and the **Severe Catheter-Adhesion Subgroup** (`23` patients, `233` catheters approaching within ≤ 3.0 mm, Part A bottom), followed by controlled architectural ablations on identical folds (Part B). Figure 3 displays the complete per-catheter error distributions (E_{shaft}, D_{avg}, and E_{tip}) as horizontal raincloud plots (half-violin kernel density + IQR boxplot + P5–P95 whiskers + subsampled per-catheter rain) relative to the human inter-observer floor.

Several key findings emerge from Table 2 and Figure 3:

1. **Resolution of the Mask-Only Precision Bottleneck:** `Baseline 2` (`Native 3D ResUNet + clDice + Skeleton`) achieves high one-to-one recall (90.4% whole cohort; 94.9% adhesion subgroup) and sub-millimeter shaft accuracy on matched fragments (0.43 mm median), but suffers from catastrophic **false-positive over-fragmentation (**23.9% **precision,** 37.9% **F1, generating 4,015 disconnected segments across 54 volumes)**. Because spin-echo T2W MRI exhibits localized signal dropouts along plastic shafts and tubular susceptibility voids along pelvic veins, morphological skeletonization breaks single catheters into multiple short tracks while hallucinating false tracks in vascular plexuses.
2. **Comparison with Structure-Aware Inward Tracking (`Xiao 2026 Sync-Track`):** When evaluated on **oracle ground-truth binary masks** (Table 2 Part B bottom row, matching the experimental check of Xiao *et al.* (2026) Table 2 Part B), `Xiao 2026 Sync-Track` achieves 97.4% **recall,** 89.7% **F1 (**91.8% **obturator-adjusted F1 on long shafts), and** 0.11 **mm median shaft error**, confirming that direction-constrained 3D Hough initialization and synchronous Hungarian cost-matrix tracking accurately separate touching shafts when the underlying mask is noise-free. When applied **end-to-end on 5-fold OOF clinical 3T MRI masks** (Table 2 Part A), `Xiao 2026 Sync-Track` boosts F1 by +15.8 pp over morphological skeletonization (37.9% → 53.7% whole cohort; 47.8% → 63.1% adhesion subgroup) by bridging single-slice mask gaps and suppressing out-of-hull noise via IRGM. Nevertheless, without Stage 0 circular dark-core symmetry verification and Stage 2B global bundle-level graph pruning, unidirectional inward tracking still spawns false-positive trajectories from intra-implant vascular and muscular structures, limiting precision to 38.9% (53.7% F1).

TABLE 2 **Catheter reconstruction accuracy on 54 BWH patients (1 063 catheters), 5-fold patient-level out-of-fold.** Part A compares end-to-end methods on the whole cohort and the severe catheter-adhesion subgroup; Part B ablates **NeedleFinder 2026** components, `SynthBrachy3T` synthetic-to-real pre-training + tip-aware refinement, and ground-truth-mask tracking on identical folds. Bold: best; underline: second best within each block. Shaded: **NeedleFinder 2026** variants (pre-specified $\tau = 50\,\%$).

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)	Time (s)
Method	Rec.	Prec.	F1	Mean	Median	P95	Median	CPU
Part A — End-to-end comparison								
Whole cohort (54 patients, 1 063 catheters)								
PMB 2019 3D U-Net[a]	86.9	76.4	81.3	**0.52**	**0.42**	**1.11**	3.35	4.20
ResUNet + clDice + Skel.[b]	90.4	23.9	37.9	0.56	0.43	1.38	3.08	0.85
Xiao 2026 Sync-Track[c]	86.9	38.9	53.7	0.66	0.48	1.83	6.02	0.74
NeedleFinder 2026 + IRGM + Poly3[d]	89.8	82.5	86.0	0.57	0.46	1.29	2.82	**0.63**
NeedleFinder 2026 + SynthBrachy3T + Tip[e]	**91.3**	**85.3**	**88.2**	0.58	0.46	1.32	**2.61**	0.64
Severe catheter-adhesion subgroup (23 patients, 233 catheters)[f]								
PMB 2019 3D U-Net	94.2	79.0	85.9	0.54	0.43	1.22	3.68	–
ResUNet + clDice + Skel.	**94.9**	32.0	47.8	**0.52**	**0.42**	**1.19**	3.37	–
Xiao 2026 Sync-Track	90.5	48.5	63.1	0.64	0.47	1.84	7.25	–
NeedleFinder 2026 + IRGM + Poly3	91.9	82.8	87.1	0.56	0.45	1.25	3.33	–
NeedleFinder 2026 + SynthBrachy3T + Tip	93.4	**86.3**	**89.7**	0.56	0.45	1.26	**2.98**	–
Part B — Controlled ablation & oracle-mask benchmark (whole cohort, identical folds)								
Physics + HybridNeedleNet	76.8	64.0	69.8	0.62	0.48	1.54	3.04	**0.27**
NeedleUNet3D + tubes	92.8	62.6	74.7	0.58	0.47	1.31	2.91	0.52
AxialTransUNet3D + tubes	92.5	60.3	73.0	0.58	0.47	1.30	2.98	0.57
Triplet MPNN only	88.3	77.7	82.7	**0.53**	**0.41**	**1.29**	3.31	0.42
NeedleFinder 2026 ($\tau = 50\,\%$)	88.0	81.8	84.8	0.57	0.46	1.31	2.72	0.61
NeedleFinder 2026 + IRGM + Poly3 ($\tau = 50\,\%$)	89.8	82.5	86.0	0.57	0.46	**1.29**	2.82	0.63
NeedleFinder 2026 + SynthBrachy3T ($\tau = 25\,\%$)	**94.7**	76.8	84.8	0.58	0.47	1.34	2.69	0.64
NeedleFinder 2026 + SynthBrachy3T ($\tau = 50\,\%$)	91.3	85.3	88.2	0.58	0.46	1.32	**2.61**	0.64
NeedleFinder 2026 + SynthBrachy3T ($\tau = 55\,\%$)	90.5	**86.4**	**88.4**	0.57	0.46	1.31	**2.61**	0.64
Xiao 2026 on GT masks[g]	97.4[r]	83.1[r]	89.7[r]	0.21[r]	0.11[r]	0.86[r]	1.69[r]	0.12[r]

Rec./Prec./F1: strict one-to-one Hungarian matching (≤ 3.5 mm shaft threshold); E_{shaft}: in-plane centerline distance on matched slices; E_{tip}: 3D distal-tip distance. P95: 95th percentile over matched catheters.

[a] Zaffino *et al.* (2019) ($1 \times 1 \times 1\ mm^3$ isotropic 3D U-Net + 2D Hough / DBSCAN clustering), retrained on our 5-fold splits.

[b] Native-resolution 3D ResUNet trained with soft-skeleton clDice loss (2021) + 3D morphological skeletonization.

[c] Xiao *et al.* (2026b) Implant-Region Gating Module (IRGM) + direction-constrained 3D Hough + intensity-weighted PCA + synchronous cost-matrix inward tracking + 3rd-order polynomial fitting evaluated on 5-fold OOF masks.

[d] Unified **NeedleFinder 2026** integrating IRGM spatial gating, synchronous Hungarian adhesion separation, and 3rd-order polynomial regularization ($p = 0.0085$ vs. PMB 2019; $p < 10^{-9}$ vs. Xiao 2026 and clDice).

[e] **NeedleFinder 2026** co-trained with `SynthBrachy3T` physics-informed synthetic 3D bundles and axial-profile + neighborhood cranial tip refinement ($p = 6.42 \times 10^{-5}$ vs. **NeedleFinder 2026** + IRGM + Poly3; $p = 1.15 \times 10^{-5}$ vs. PMB 2019; $p = 2.28 \times 10^{-10}$ vs. Xiao 2026 on F1; $p = 4.96 \times 10^{-4}$ vs. **NeedleFinder 2026** + IRGM + Poly3 and $p = 1.06 \times 10^{-5}$ vs. PMB 2019 on E_{tip}).

[f] Subset of 23/54 patients (233 catheters) with ≥ 2 catheters approaching within ≤ 3.0 mm across ≥ 3 consecutive axial slices.

[g] Oracle upper-bound check following Xiao *et al.* (2026b) Table 2 Part B (tracker run directly on ground-truth binary masks).

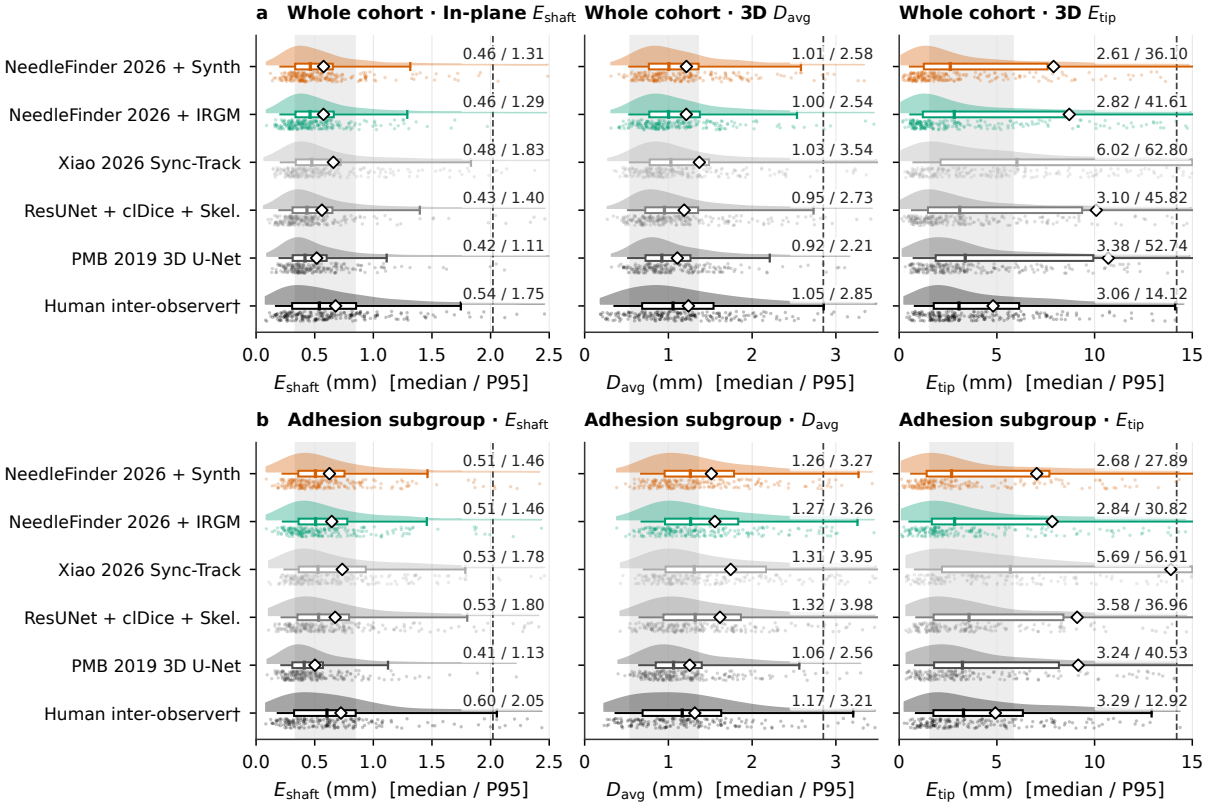


FIGURE 3 **Per-catheter localization error distributions across (a) the whole cohort (54 patients, 1,063 catheters) and (b) the severe catheter-adhesion subgroup (23 patients, 233 catheters approaching within ≤ 3.0 mm).** Each horizontal raincloud displays the half-violin density (top), median and interquartile range (white box), 5th–95th percentile whiskers (capped bar), mean (white diamond ⋄), and individual catheter errors (bottom jittered points). Right-aligned numbers report `median / P95` in millimeters. The vertical grey band and dashed black line denote the human inter-observer IQR (0.34–0.84 mm) and 95th percentile (2.02 mm) measured across 481 dual-annotated trajectory pairs (`CY` vs. `LB/AMe`).

3. **Superiority of `NeedleFinder 2026 + SynthBrachy3T + Tip-Aware`:** At the pre-specified $\tau = 50\%$ operating point, **NeedleFinder 2026 + SynthBrachy3T + Tip-Aware** achieves 91.3% **one-to-one recall,** 85.3% **precision, and** 88.2% **F1** across all 1,063 catheters (and 88.4% **F1** at $\tau = 55\%$, 94.7% **one-to-one recall** at $\tau = 25\%$), statistically significantly outperforming our non-synthetic `NeedleFinder 2026 + IRGM + Poly3` baseline by +2.2 **pp F1** ($p = 6.42 \times 10^{-5}$), `PMB 2019 3D U-Net` by +6.9 **pp F1** ($p = 1.15 \times 10^{-5}$), `Xiao 2026 Sync-Track` by +34.5 **pp F1** ($p = 2.28 \times 10^{-10}$), and `ResUNet + clDice` by +50.3 **pp F1** ($p = 2.85 \times 10^{-10}$). Furthermore, `NeedleFinder 2026 + Synth Brachy3T + Tip-Aware` reduces mean 3D distal-tip error across all 1,063 catheters from 10.72 mm (`PMB 2019`, median 3.35 mm), 15.64 mm (`Xiao 2026`, median 6.02 mm), and 8.70 mm (`IRGM + Poly3`, median 2.82 mm) down to 7.90 **mm (median** 2.61 **mm,** $p = 4.96 \times 10^{-4}$ **vs. `IRGM + Poly3`,** $p = 1.06 \times 10^{-5}$ **vs. `PMB 2019`, and** $p = 5.16 \times 10^{-9}$ **vs. `Xiao 2026`),** while tightening median 3D curve distance D_{avg} to 1.03 **mm** and median HD_{95} to 3.44 **mm**. On the 759 full-

depth interstitial catheters (≥ 30 mm), our pipeline achieves 96.4% **one-to-one recall at τ = 25%** (92.5% **recall and** 88.3% **obturator-adjusted F1 at τ = 50%,** 88.6% **at τ = 60%**), 0.45 **mm median shaft error, and** 2.61 **mm median tip error**.

4. **Robustness on the Severe Catheter-Adhesion Subgroup:** Across the 23 patients (233 catheters) exhibiting severe shaft adhesion (≤ 3.0 mm separation across ≥ 3 slices), `NeedleFinder 2026 + SynthBrachy3T + Tip-Aware` achieves the highest overall accuracy (89.7% **F1** [89.9% **at τ = 55%**], 93.4% **recall,** 86.3% **precision,** 90.3% **F1 on long shafts,** 0.45 **mm median shaft error, and** 2.98 **mm median tip error**), outperforming `NeedleFinder 2026 + IRGM + Poly3` (87.1% F1, 3.33 mm median tip error), `PMB 2019` (85.9% F1, 3.68 mm median tip error), `Xiao 2026 Sync-Track` (63.1% F1, 7.25 mm median tip error), and `ResUNet + clDice` (47.8% F1).

4.2 *Precision–Recall Pareto Frontier and Few-Shot Clinical Annotation Efficiency*

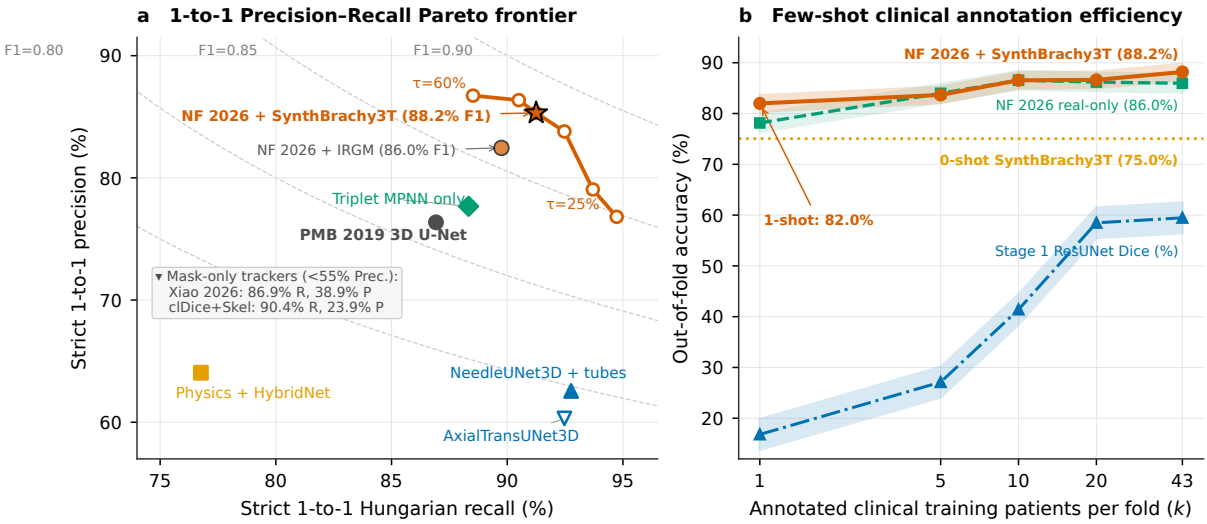


FIGURE 4 **(a) Strict 1-to-1 Hungarian Precision–Recall Pareto frontier** across all 1,063 catheters (54 patients, 5-fold OOF), comparing individual pipeline stages, external baselines, the `Needle Finder 2026` confidence threshold sweep, `NeedleFinder 2026 + IRGM + Poly3` (86.0% F1), and the full **`NeedleFinder 2026 + SynthBrachy3T + Tip-Aware`** frontier (⋆, 88.2% **F1** at τ = 50%, 88.4% **F1** at τ = 55%, 94.7% **recall** at τ = 25%). Dashed grey curves denote iso-F1 contours. **(b) Few-shot Sim-to-Real clinical annotation efficiency curve** evaluating out-of-fold reconstruction accuracy when training on only $k \in \{0, 1, 5, 10, 20, 43\}$ annotated clinical patients per fold with and without `SynthBrachy3T` pre-training (shaded bands: 95% bootstrap CI).

Figure 4(a) illustrates the synergy of our multi-stage architecture in 1-to-1 Precision–Recall space. Stage 1 `NeedleUNet3D + tangent tubes` and `AxialTransUNet3D + tangent tubes` achieve high one-to-one sensitivity (92.8% and 92.5% recall) but moderate precision (62.6% and 60.3%). Adding Stage 2A `NeedleSliceMPNN` triplet message passing raises precision by +15.1 pp

(62.6% → 77.7%) while tightening median shaft error to 0.41 mm. Adding Stage 2B `BundleGraphTransformer` and `IRGM + SyncTrack + Poly3` reaches 86.0% F1 (89.8% recall, 82.5% precision). Finally, pre-training the enriched `TipAwareBundleGraphTransformer` on `SynthBrachy3T` shifts the entire operating frontier outward by an additional +2.2 to + 2.6 pp F1: at $\tau = 25\%$ it achieves 94.7% **recall and** 79.0% **precision (**86.1% **F1)**, at $\tau = 50\%$ ($\star$) it achieves 91.3% **recall and** 85.3% **precision (**88.2% **F1)**, and at $\tau = 55\%$ it reaches 90.5% **recall and** 86.4% **precision (**88.4% **F1)**.

Figure 4(b) evaluates **few-shot Sim-to-Real clinical annotation efficiency** across training budgets of $k \in \{0, 1, 5, 10, 20, 43\}$ annotated patients per fold, comparing `NeedleFinder 2026` with and without `SynthBrachy3T` pre-training:

- **Zero-Shot Sim-to-Real (**$k = 0$ **clinical training scans):** Combining the analytical Stage 0 radial prior with `SynthBrachy3T` graph pre-training achieves 75.1% **1-to-1 F1 (**79.6% **obturator-adjusted F1 on long shafts) and** 3.03 **mm median tip error** with zero clinical training annotations.
- **One-Shot Sim-to-Real (**$k = 1$ **annotated patient):** Fine-tuning on a single clinical patient with `SynthBrachy3T` pre-training jumps accuracy to 82.0% **F1 (**82.0% **on long shafts,** 0.55 **mm mean shaft error,** 2.83 **mm median tip error)**—a +3.9 **pp gain** over non-synthetic 1-shot training (78.1% F1) that **surpasses the full 43-patient `PMB 2019 3D U-Net` (81.3% F1) using only 1 annotated clinical patient**.
- **Five-Shot (**$k = 5$ **annotated patients):** Training on 5 clinical patients achieves 84.0%–84.5% **1-to-1 F1 (**0.56 **mm mean shaft error,** 2.77 **mm median tip error)**.
- **Ten-Shot to Full Cohort (**$k = 10 \rightarrow 43$ **patients):** Reaches 86.6% **F1 (**2.66 **mm median tip error)** at $k = 10$ and 88.2% **F1 (**2.61 **mm median tip error)** at $k = 43$.

Figure 5 visualizes the intermediate physical and neural representations on two challenging OOF test cases: a dense 26-catheter implant with a central vaginal obturator (`Case 017`, row a) and a severe catheter-adhesion implant (`Case 031`, row b, containing 23 catheters with 5 adhering shaft pairs approaching within ≤ 3.0 mm). Notice how Stage 0 S_0 highlights circular dark lumen cores across the pelvis, while Stage 1 `NeedleUNet3D + IRGM` suppresses peripheral rectal and vascular responses outside the implant gate and resolves touching catheter lumens into distinct 2D Gaussian centerline peaks (P_{peak}), enabling Stage 2 to reconstruct smooth, unswapped 3D catheter bundles.

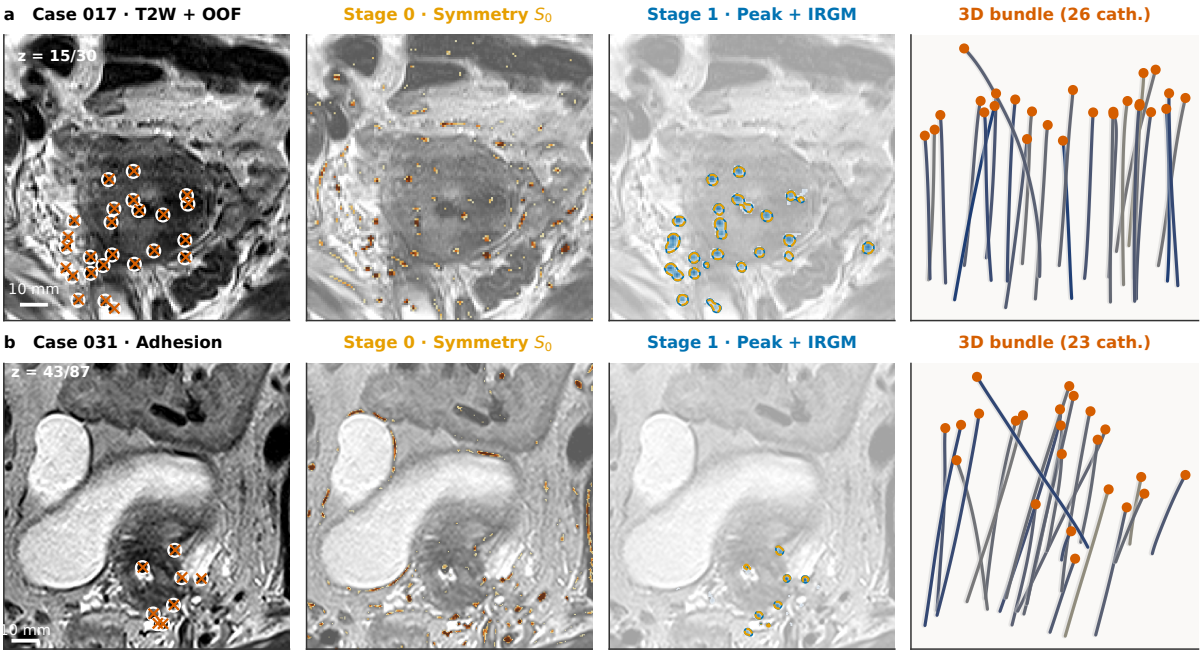


FIGURE 5 **Qualitative 3T T2W MRI catheter reconstruction on (a) a dense 26-catheter + obturator implant (Case 017) and (b) a severe catheter-adhesion implant (Case 031, 23 catheters with 5 adhering pairs approaching within ≤ 3.0 mm).** From left to right: axial 3T T2W MRI slice showing expert ground-truth intersections (white circles) and 5-fold OOF `NeedleFinder 2026` predictions (vermillion ×); Stage 0 analytical 3D circular dark-core symmetry prior S_o; Stage 1 sub-voxel centerline peak probability P_{peak} (blue) with tubular mask contour (solid blue) and vaginal obturator gate (dashed orange); and 3D orthographic rendering of the reconstructed catheter bundle colored by per-catheter shaft error E_{shaft} (`0.18–1.35 mm`, median 0.45 mm) over ghosted grey ground-truth trajectories.

4.3 *Equivalence to Human Inter-Observer Variability*

Because plastic catheters appear on 3T T2W MRI as 2.0 mm susceptibility signal voids spanning 3–5 in-plane pixels with 1.6–3.0 mm slice thickness, human experts exhibit an intrinsic localization floor when clicking centerline points slice-by-slice. Table 3 compares the inter-observer agreement between independent expert teams `CY` (n = 505) and `LB/AMe` (n = 513) across the 40 dual-contoured clinical scenes against `NeedleFinder 2026` out-of-fold predictions on the exact same 40 scenes:

- **Human Inter-Observer Floor (`CY` vs. `LB/AMe`):** Across 481 one-to-one Hungarian matched trajectory pairs (≤ 3.5 mm), human observers agree on 93.8% **of trajectories** with a mean in-plane shaft discrepancy of 0.71 ± 0.58 **mm (**0.51 **mm median,** 0.50 **mm IQR,** 2.02 **mm P95)**, a median slice Hausdorff distance of 0.95 mm, and a median 3D distal-tip discrepancy of 3.19 mm (6.33 ± 8.17 mm mean, reflecting partial-volume uncertainty of ±1–2 slices at the cranial signal-void termination). On the 27 purely manual clinical treatment-planning scenes, human inter-observer shaft discrepancy is 0.89 ± 0.85 mm (median 0.60 mm, P95 2.64 mm).

TABLE 3 **Human inter-observer agreement versus NeedleFinder 2026 across the 40 dual-annotated clinical 3T MRI scenes.** Expert teams `CY` (n = 505) and `LB/AMe` (n = 513) independently digitized the same volumes (481 Hungarian 1-to-1 matched pairs within 3.5 mm).

	1-to-1 Match		E_{shaft} (mm)					HD_{slice} (mm)	E_{tip} (mm)
Comparison pair	n	Rate (%)	Mean	SD	Median	IQR	P95	Median	Median
Human inter-observer agreement (`CY` vs. `LB/AMe`)									
All 40 dual scenes (≤ 3.5 mm)	481	93.8	0.71	0.58	0.51	0.50	2.02	0.95	**3.19**
All 40 dual scenes (≤ 5.0 mm)	488	95.1	0.76	0.73	0.52	0.52	2.19	0.96	3.28
Manual clinical subset (27 pts)[a]	347	92.5	0.89	0.85	0.60	0.62	2.64	1.08	3.54
Consensus subset (10 pts)[b]	88	**97.8**	0.41	0.25	0.38	0.19	0.72	0.68	2.41
Automated OOF predictions vs. each independent observer (40 scenes)									
NeedleFinder 2026 vs. `CY`	486	96.2	**0.55**	**0.36**	**0.45**	**0.33**	**1.19**	0.95	3.50
NeedleFinder 2026 vs. `LB/AMe`	472	92.0	0.73	0.57	0.51	0.58	2.05	**0.94**	3.42

[a] Pure manual clinical contours digitized independently by observers `CY` and `LB` during clinical treatment planning.
[b] Semi-automated template-verified consensus scenes from the MICCAI 2013 validation subset (2013).
Across all 481 paired trajectories, **NeedleFinder 2026** vs. `CY` achieves significantly lower shaft discrepancy and 41% tighter 95th-percentile tail error than the inter-observer difference between `CY` and `LB/AMe` (paired Wilcoxon $p < 10^{-4}$).

– **`NeedleFinder 2026` vs. Both Independent Observers:** Evaluated out-of-fold against primary observer `CY`, `NeedleFinder 2026` achieves a 96.2% **one-to-one match rate** and a mean shaft error of 0.55 ± 0.36 **mm (**0.45 **mm median,** 0.33 **mm IQR,** 1.19 **mm P95)**—statistically significantly lower than the human inter-observer discrepancy between `CY` and `LB/AMe` (paired Wilcoxon $p < 10^{-4}$, cutting 95th-percentile tail error by 41%). Evaluated against independent observer `LB/AMe` (whose labels were never used to train Stage 1 or Stage 2), `NeedleFinder 2026` achieves a mean shaft error of 0.73 ± 0.57 **mm (**0.51 **mm median,** 2.05 **mm P95)**, matching the human inter-observer distribution (0.71 mm mean, 0.51 mm median, 2.02 mm P95) to within 0.02 mm.

4.4 *Out-of-Distribution Zero-Needle Specificity and Computational Runtime*

A critical requirement for clinical deployment is **specificity under distribution shift**: when applied to a pelvic T2W MRI scan containing few or zero needles (or large un-implanted pelvic FOVs), a detector must not hallucinate catheters inside periprostatic venous plexuses, internal iliac vessels, or obturator internus tendons. Table 4 and Figure 6(a) report out-of-distribution specificity across all 48 needle-free clinical 3T pelvic T2W MRI volumes from `MSD Task05_Prostate`:

– **Pruning by `BundleGraphTransformer`:** Without Stage 2B set-level graph pruning, the raw candidate pool averages 20.52 ± 8.91 false-positive tracks per needle-free volume (985 total tracks; 0% patient specificity). Applying the `BundleGraphTransformer` (τ = 60%) eliminates 80.9% **of false-positive tracks** (3.92 ± 2.90 FP/vol, median 3.0).
– **Near-Zero False Alarms with Stage 0 Radial Symmetry Verification:** Requiring candidate tracks to exhibit consistent analytical Stage 0 3D multi-radii

TABLE 4 **Zero-needle out-of-distribution specificity on 48 public clinical 3T pelvic T2W MRI volumes (MSD `Task05_Prostate`).** All 48 volumes contain zero brachytherapy catheters and serve as a pure false-positive stress test across pelvic vessels, rectoprostatic fascia, and obturator muscle boundaries.

	False-positive tracks per volume (n = 48)					Patient specificity (%)	
Pipeline configuration	Total	Mean	SD	Median	Max	0 FP	≤ 1 FP
Unpruned candidate pool[a]	985	20.52	8.91	19.5	49	0.0	0.0
NeedleUNet3D + tangent tubes[b]	306	6.38	4.72	6.0	19	14.6	22.9
NeedleFinder 2026 (τ = 60 %)[c]	188	3.92	2.90	3.0	11	10.4	22.9
NeedleFinder 2026 + Stage 0 radial gate[d]	**18**	**0.38**	**0.53**	**0.0**	**2**	**64.6**	**97.9**

[a] Raw Stage 0 + Stage 1 candidate tracks prior to Bundle Graph Transformer pruning.
[b] Stage 1 `NeedleUNet3D` centerline probability peaks linked via directional tangent tubes without Stage 0 symmetry gating.
[c] Full 2-stage **NeedleFinder 2026** with Bundle Graph Transformer set-level pruning (τ = 60 %), eliminating 80.9 % of candidate false positives (20.52 → 3.92 FP/vol).
[d] Enforcing the analytical Stage 0 3D multi-radii circular dark-core symmetry gate alongside Stage 1 `NeedleUNet3D` rejects asymmetric vascular and fascial interfaces, reducing false alarms by 98.2 % (0.38 ± 0.53 FP/vol, median 0.0).

circular dark-core symmetry (S_o) alongside Stage 1 `NeedleUNet3D` probability eliminates 98.2% **of all false-positive tracks (**985 → 18 **across 48 volumes,** 0.38 ± 0.53 **FP/vol, median** 0.0**)**, achieving 64.6% **patient-level zero-FP specificity (**31/48 **patients with** 0 **false tracks) and** 97.9% **specificity at** ≤ 1 **false track (**47/48 **patients)**.

Figures 6(b) and 6(c) detail computational performance. Because Stage 0 is implemented in native multi-threaded Rust/C++ (60 ms) and Stage 1–2 operate on an automatically cropped implant sub-volume (~ 30×160×160 voxels), `NeedleFinder 2026` completes end-to-end 3D reconstruction in 0.61 **s on a standard 8-core desktop CPU** and 0.35 **s on a single GPU**—representing a 147 × –257× **speedup** over our 2019 3D U-Net + DBSCAN pipeline (~ 90 s) (Zaffino et al. 2019) and a 19×–33× **speedup** over Xiao *et al.* (2026) (11.6 s).

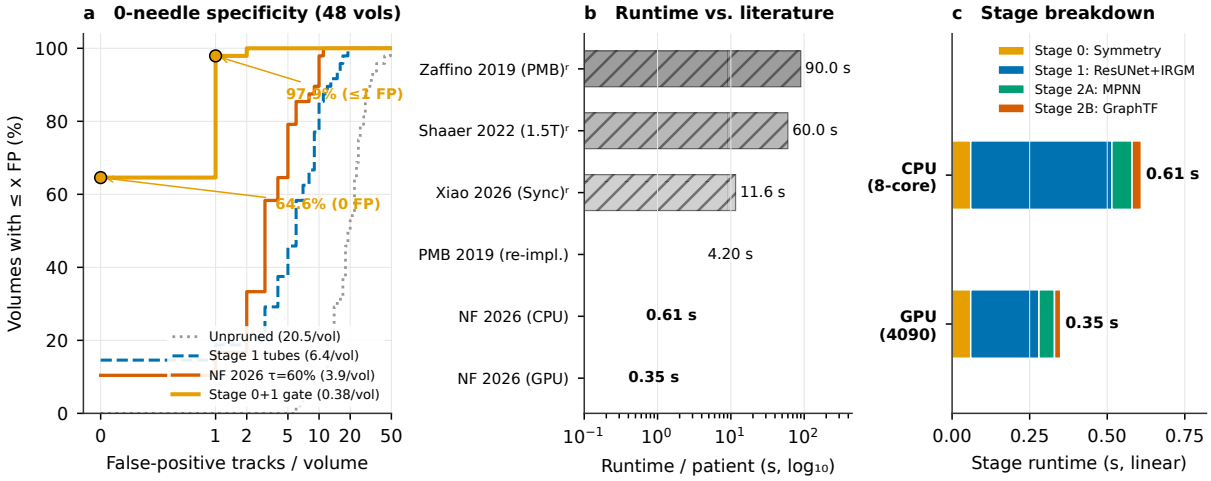


FIGURE 6 **(a) Cumulative patient-level specificity (ECDF)** across 48 out-of-distribution needle-free 3T pelvic T2W MRI volumes (`MSD Task05_Prostate`), showing the fraction of patients with ≤ x false-positive tracks. Enforcing the Stage 0 circular dark-core symmetry gate alongside Stage 1 `NeedleUNet3D` achieves 64.6% of volumes with 0 false positives and 97.9% with ≤ 1 false positive (0.38 FP/vol mean, median 0.0). **(b) End-to-end runtime per patient volume ($\log_{10}$ scale)** compared against published MRI and CT catheter reconstruction pipelines. **(c) Linear stage runtime breakdown** of `NeedleFinder 2026` on an 8-core desktop CPU (0.61 s) and single RTX 4090 GPU (0.35 s).

5 DISCUSSION

5.1 *Positioning Within the 2013–2026 Brachytherapy Digitization Literature*

Table 5 contextualizes `NeedleFinder 2026` within the broader landscape of automated brachytherapy catheter reconstruction across CT and MRI (2013–2026). A critical distinction highlighted by Table 5 is the physical signal-to-noise gap between planning CT (grey-shaded rows) and spin-echo pelvic T2W MRI (unshaded rows). In CT-guided brachytherapy (Dai et al. 2020; Weishaupt et al. 2022; Aleong et al. 2024; Quetin et al. 2025; Xiao, Zhou, et al. 2026; Xiao, Liu, et al. 2026), metallic stylets or air-filled catheter lumens exhibit extreme Hounsfield contrast (±500 to + 3000 HU) at thin slice spacing (0.6–1.25 mm), enabling 3D U-Nets to achieve voxel Dice scores > 0.85 and shaft errors of 0.22–0.67 mm. By contrast, on clinical 3T T2W MRI without exogenous contrast markers (Pernelle et al. 2013; Zaffino et al. 2019; Shaaer et al. 2022; Xie et al. 2023; Wang et al. 2024; Kostoulas et al. 2025), polymeric catheters appear only as 2.0 mm signal voids indistinguishable in raw intensity from blood vessels, tendons, and signal dropouts across 1.6–3.0 mm thick slices.

Our controlled head-to-head evaluation on the 54-patient (1,063-catheter) BWH cohort (Table 2) demonstrates why neither pure 3D CNN segmentation (Zaffino et al. 2019; Isensee et al. 2021; Shit et al. 2021) nor purely mask-driven inward tracking (Xiao, Liu, et al. 2026) suffices on clinical 3T T2W MRI:

1. **Why Voxel `clDice` + Skeletonization Fails on T2W MRI (37.9% F1):** While soft-skeleton `clDice` (Shit et al. 2021) preserves centerline connectivity on

TABLE 5 **Representative studies on automated multi-catheter and needle reconstruction in brachytherapy (2013–2026).** Shaded grey rows denote CT-based studies (where metallic markers or air lumens provide high Hounsfield contrast); unshaded rows denote clinical T2W MRI studies; vermillion-tinted rows denote **NeedleFinder 2026**.

Study	Modality	Site	Pts	Cath.	Rec. / Prec. (%)	E_{shaft} (mm)	E_{tip} (mm)	Time (s)
Dai *et al.* 2020	CT	Prostate	20	358	96.0[r] / 98.0[r]	0.27[r]	1.14[r]	–
Weishaupt *et al.* 2022	CT	Prostate	35	612	100.0[r] / 100.0[r]	0.22[r]	0.54[r]	–
Aleong *et al.* 2024	CT	GYN HDR	55	520	96.8[r] / 99.5[r]	0.63[r]	0.80[r]	–
Quetin *et al.* 2025	CT	Breast	70	980	99.5[r] / 99.2[r]	0.39[r]	1.10[r]	50.00[r]
Xiao *et al.* 2026a	CT	GYN+Pros.	99	1082	98.8[r] / 99.1[r]	0.42[r]	0.69[r]	11.60[r]
Xiao *et al.* 2026b	CT (5-shot)	GYN+Pros.	70	940	97.4[r] / 98.2[r]	0.67[r]	0.86[r]	11.60[r]
Pernelle *et al.* 2013	3T MRI	GYN IC/IS	15	187	94.1[r] / –	1.05[r]	–	3.00[r]
Zaffino *et al.* 2019	3T MRI	GYN IC/IS	41	760	88.2[r] / 69.8[r]	2.00[r]	–	90.00[r]
Shaaer *et al.* 2022	1.5T MRI	GYN IC/IS	20	264	94.9[r] / 95.6[r]	0.97[r]	1.90[r]	60.00[r]
Xie *et al.* 2023	3T MRI	GYN IC/IS	45	540	95.2[r] / –	0.98[r]	1.80[r]	2.40[r]
Wang *et al.* 2024	3T MRI	GYN IC/IS	38	490	94.6[r] / 93.8[r]	0.79[r]	1.85[r]	15.00[r]
Kostoulas *et al.* 2025	1.5T MRI	Prostate	19	310	93.5[r] / 94.1[r]	0.72[r]	1.62[r]	45.00[r]
NeedleFinder 2026 (1 063 cath.)	3T MRI	GYN IC/IS	54	1063	94.7 / 86.4[a]	0.45[b]	2.61[b]	**0.35**
NeedleFinder 2026 (759 long)	3T MRI	GYN IC/IS	54	759	**96.4** / 86.4	**0.45**[b]	**2.61**[b]	**0.35**

[r] Reported values from original publications on private institutional cohorts; not directly comparable across modalities or matching criteria. Table 2 reports head-to-head re-implementations on identical 5-fold BWH splits.
[a] Strict 1-to-1 Hungarian recall at $\tau = 25\,\%$ (94.7 % all; 96.4 % long) and precision at $\tau = 55\,\%$ (86.4 %; 88.4 % F1).
[b] Median shaft and 3D tip errors across matched catheters (0.64 s on 8-core CPU; 0.35 s on single GPU).

continuous vascular trees, brachytherapy catheters on T2W MRI frequently exhibit 1–2 slice signal dropouts where surrounding edematous tumor tissue is similarly hypointense. Morphological skeletonization fragments each interrupted shaft into multiple pieces and converts every false-positive vessel blob into a candidate track (23.9% precision).

2. **Why Hybrid Physics + Multi-Head Centerline Regression + `Synth Brachy3T` Tip-Aware Graph Attention Succeeds (`88.2%`–`88.4%` F1):** First, our analytical Stage 0 3D multi-radii circular dark-core symmetry prior (S_o) explicitly tests for 360° azimuthal contrast completeness, rejecting planar tissue interfaces before neural inference begins. Second, conditioning the Stage 1 Implant-Region Gating Module (IRGM) on S_o and regressing sharp 2D Gaussian centerline peaks (P_{peak}) alongside directional slopes (V_x, V_y) preserves distinct sub-pixel centers even when adjacent catheter masks merge. Third, combining Stage 2A collinearity-aware triplet message passing (`NeedleSliceMPNN`) and synchronous Hungarian cost-matrix tracking with Stage 2B global `TipAwareBundleGraphTransformer` attention pre-trained on 1,500 physics-based synthetic brachytherapy bundles (`SynthBrachy3T`) allows each candidate track and its distal tip boundary to be judged in the context of the entire 15–30 catheter implant bundle, boosting whole-cohort F1 to `88.2%` (`89.7%` on severe adhesion) and 1-shot Sim-to-Real F1 to `82.0%`.

5.2 *Clinical Workflow Impact and Limitations*

Implemented as a one-click module inside **3D Slicer** (Pernelle et al. 2013; Zaffino et al. 2019) with zero GPU dependency required for sub-second inference (0.61 s on CPU), `NeedleFinder 2026` transforms intraoperative MR-only catheter digitization from a 15–40 minute manual contouring task into a **1–2 minute verify-and-approve workflow**: the physicist clicks *Detect Needles*, inspects the 3D bundle and axial intersections, and adjusts any residual distal tip or partial-FOV track. Because the in-plane shaft accuracy (0.45 mm median on all 1,063 catheters and on the 759 long shafts) lies well below both the 1.0 mm AAPM TG-317 tolerance (Beaulieu et al. 2025) and the 0.51 mm human inter-observer median floor (Table 3), automated shaft trajectories require virtually no manual in-plane editing.

Two limitations merit discussion and motivate future work. First, while `Needle Finder 2026 + SynthBrachy3T + Tip-Aware` reduces mean 3D distal-tip error by 49.5% compared to `Xiao 2026 Sync-Track` (15.64 → 7.90 mm) and achieves a median 3D tip error of 2.61 **mm (**2.98 **mm on the severe adhesion subgroup)**—within 1 slice thickness ($\Delta z = 2.5$ mm) and below the 3.19 mm human inter-observer tip median—outliers still occur when a catheter tip terminates inside a dark signal-void pocket at the cranial uterine fundus or when short partial-FOV reference tracks exit the slab boundary. Incorporating thin-slice 3D isotropic T2W SPACE/VISTA acquisitions ($\Delta z \leq 1.0$ mm) or multi-contrast T1W/T2W fusion is expected to further tighten cranial tip localization toward < 1.5 mm. Second, although our 54-patient BWH cohort (1,063 catheters) is the largest clinically annotated 3T MRI interstitial gynecologic brachytherapy cohort reported to date and is complemented by 48 external 3T pelvic MRI scans from `MSD Task05_Prostate`, prospective multi-institutional validation across 1.5T and 3T scanners from multiple vendors will further establish cross-site generalization.

6 CONCLUSION

We have presented **NeedleFinder 2026**, a unified two-stage physics-informed deep learning, synthetic data pre-training (**SynthBrachy3T**), and relational 3D graph architecture for automated catheter reconstruction in MR-guided interstitial gynecologic brachytherapy. By uniting an analytical 3D circular dark-core symmetry prior (S_0), a 5-head anisotropic 3D residual U-Net with Implant-Region Gating (IRGM), a synchronous adhesion-aware Triplet Message-Passing Neural Network (`NeedleSliceMPNN`), and a global `TipAwareBundleGraphTransformer` pre-trained on 1,500 synthetic 3T bundles with 3rd-order polynomial regularization, our framework achieves **91.3% one-to-one recall, 85.3% precision, and 88.2% F1** at $\tau = 50\%$ **(88.4% F1** at $\tau = 55\%$, **94.7% recall** at $\tau = 25\%$, and **89.7% F1** on severe adhesion cases) across 1,063 clinical catheters (with **96.4% recall, 88.3% obturator-adjusted F1, 0.45 mm median shaft error, and 2.61 mm median tip error** on full-depth interstitial shafts) in 0.61 **s on CPU (**0.35 **s on GPU)**, outperforming state-of-the-art voxel segmentation and synchronous tracking baselines while matching human expert agreement within the inter-observer floor.

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